

Women Safety App - Detailed Guide

Generated: 2025-12-19

1) Purpose and Core Value

- Rapid SOS with multi-channel alerts, real-time location sharing, AI safety checks, and a one-tap panic widget.
- Built with Flutter (Dart), Firebase backend, Kotlin widget bridge, and a modular service architecture.

2) User-Facing Features (How they work)

- SOS Flow: User taps SOS → countdown → loud alarm → evidence capture (photo/audio/video optional) → alerts via WhatsApp/SMS/Email/Push → call escalation to guardians.
- Location Sharing: From Map screen or Profile Quick Action, fetches current GPS (Geolocator), builds Google Maps link, lets user pick guardians, and opens WhatsApp/SMS with prefilled message.
- Guardians: Add/edit/remove guardians, mark primary. Stored in Firestore. Used by SOS and share flows to target contacts.
- Revolutionary Features Hub (8 items):
 - 1) AI Danger Prediction (TFLite model) - risk score based on context.
 - 2) Voice Distress Analysis - detects distress patterns/keywords from mic.
 - 3) Face Recognition - trusted faces vs unknown using ML Kit.
 - 4) Fake Call - schedules simulated incoming call UI.
 - 5) Safe Routes - opens Google Maps walking directions from current location.
 - 6) Evidence Capture - photo/audio/video for proof.
 - 7) Live Streaming (Agora) - realtime video to guardians (needs APP_ID).
 - 8) Guardian Volunteers - community/extended guardian network.
- Ride/Route Safety: Ride tracking (Uber/Ola), safe journey monitoring, route deviation alerts, shake-to-SOS, voice activation trigger.
- Panic Home Widget: 2x2 Android widget with red SOS button, status text, contact count; tapping triggers panic intent or launches app.
- Settings & Profile: Theme toggle, notification and permission prompts, password reset, Quick Actions section.

3) Key Screens and Navigation

- Routes defined in lib/app.dart: Splash (/), Login (/login), Home (/home), SOS (/sos), Map (/map), Profile (/profile), Settings (/settings), Revolutionary Features (/revolutionary-features), Guardian Management (/guardians), Safety Checklist (/safety-checklist).
- Profile Quick Actions: Share My Location → /map; Revolutionary Features → /revolutionary-features; Panic Widget Setup dialog.
- Map Screen: OpenStreetMap view, current location, Share button (WhatsApp/SMS contacts dialog).
- Revolutionary Features: Lists 8 advanced items; Safe Routes FAB opens Google Maps directions.

4) Core Services (lib/services) and Responsibilities

- sos_service.dart: Orchestrates SOS steps; coordinates alarm, evidence, alerts, escalation.
- location_share_service.dart: Gets GPS, builds Maps link, shares via WhatsApp/SMS with fallbacks.
- whatsapp_service.dart: Formats SOS messages, per-contact send with delay and error handling.
- sms_service.dart: Compact SOS text, multiple URI schemes (sms:, smsto:, mms:) to maximize compatibility.
- panic_widget_service.dart: Bridges Flutter and Android widget (HomeWidget storage, trigger checks, updates).
- protection_service.dart: Background pending SOS trigger checks.
- notification_service.dart: Local/push notification helper APIs.
- permissions_service.dart: Centralized runtime permission requests (location, notifications, camera, mic).
- ai_danger_prediction_service.dart: TFLite model inference for risk.
- distress_voice_analysis_service.dart: Audio-based distress detection.
- face_recognition_service.dart: Face match/unknown detection using ML Kit.
- live_streaming_service.dart: Agora integration for video; requires APP_ID config.
- offline_queue_service.dart: Queues actions when offline and retries when online.
- guardian_network_service.dart / guardian_tracking_service.dart / ride_tracking_service.dart: Guardian management and journey safety.
- fake_call_service.dart: Simulated incoming call UI.

5) Android Widget Implementation

- PanicWidgetProvider.kt: AppWidgetProvider handling updates and PANIC_BUTTON_CLICKED intent; reads

HomeWidget data (user_name, contact_count, status).

- Layout: res/layout/panic_widget.xml (red gradient card, circular SOS button, status text, contact count).
- Metadata: res/xml/panic_widget_info.xml (min size, category home_screen, preview).
- Drawables: widget_background.xml (gradient), panic_button_bg.xml (circular red), panic_widget_preview.xml.
- Manifest: Receiver registered with APPWIDGET_UPDATE and custom panic intent.

6) Startup and Flows (main.dart)

- Initializes Firebase, notification service, permission checks, offline queue, panic widget.
- Checks ProtectionService pending triggers and PanicWidgetService widget trigger.
- Provides blocs (ThemeCubit, AuthCubit, LocationCubit, SosCubit) and runs WomenSafetyApp with navigatorKey.

7) Data and Models

- app_user.dart: user profile basics.
- guardian.dart: emergency contact info, primary flag.
- sos_alert.dart: alert payload structure.
- Firestore collections: users (with emergencyContactIds), guardians, alerts/logs (as configured).

8) Setup Requirements

- Firebase: Add android/app/google-services.json (critical). iOS plist if applicable.
- Permissions: Location, camera, microphone, notifications, vibration; accept runtime prompts.
- Optional: Agora APP_ID in live_streaming_service.dart; TFLite model in assets/models/ for danger prediction.
- Assets: Ensure animations/models folders present if using Lottie/TFLite.

9) Testing Checklist (manual)

- Profile → Quick Actions → all 3 entries work (Map, Features, Widget dialog).
- Map → Share → choose guardian → WhatsApp/SMS opens with Maps link.
- Revolutionary Features → Safe Routes FAB opens Google Maps.
- Widget → Add to home (launcher supports widgets) → tap triggers app/intent.
- SOS → triggers alerts (with real creds, permissions granted).

10) Common Issues and Fixes

- Firebase init fails: Add google-services.json to android/app.
- WhatsApp/SMS not opening: Ensure app installed and phone numbers include country code.
- Location fails: Grant location permission in system settings.
- Widget missing: Some launchers hide widgets; ensure category=home_screen and try a standard launcher.

11) Build and Install

- Debug: flutter run -d android
- Release APK: flutter build apk --release
- Install via ADB: adb install -r build/app/outputs/flutter-apk/app-release.apk

For a richer visual doc, open project_documentation.html in the same folder.